Automated Irregular Shaped Stone Stacking via Pose Optimization and GPU-accelerated Monte Carlo Tree Search

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***Abstract*—Stacking irregular objects into prescribed target shapes without container walls requires mutually supporting arrangements despite uneven contact surfaces. The problem couples combinatorial object selection and ordering with continuous pose optimization, as each placement redistributes loads and changes support geometry and space available for subsequent objects. We propose a framework with two complementary components: structure-aware bilevel pose optimization and parallel Monte Carlo Tree Search (MCTS). The optimizer combines target-shape objectives with contact-force optimization to account for force and torque equilibrium across the structure under friction constraints, using bodywise ADMM updates for GPU parallelization. The receding-horizon MCTS planner evaluates batches of candidate placements in parallel through pose refinement and physical simulation, providing lookahead over object choices and placement poses. We hypothesize that combining structural stability evaluation with lookahead improves target coverage while maintaining stability under a fixed planning budget.**

I. Introduction

Stacking is a fundamental problem with applications in bin packing, industrial warehouse storage [20, 9], and stone-wall construction [12]. In freestanding stacking, the resulting structure must approximate a prescribed target shape while maintaining force and torque equilibrium under contact forces and gravity, without support from an enclosing container. For n irregular objects, this geometric goal can be expressed as minimizing the unfilled volume of a target shape $T \subset \mathbb{R}^3$, subject to physical feasibility:

$$\min_{q_1,\cdots,q_n \in \mathrm{SE}(3)} \mathbb{V}\left(T \setminus \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \quad c_{\text{phys}}(O_1(q_1),\cdots,O_n(q_n)) = 0, \tag{1}$$

where $O_i(q_i) \subset \mathbb{R}^3$ is the region occupied by object i at pose $q_i \in \mathrm{SE}(3)$, $\mathbb{V}(\cdot)$ denotes volume, and $c_{\text{phys}} = 0$ represents stability and non-interpenetration constraints. Achieving these requirements simultaneously is challenging because changing one object pose alters both the occupied space and the contact geometry through which loads are transmitted across the structure. Moreover, the choice and pose of lower objects constrain the remaining feasible subsequent placements, so even a currently stable arrangement may limit further stacking toward the target shape, especially when the available object inventory is limited. Planning its construction therefore involves combinatorial choices of objects and placement order, together with iterative numerical evaluation of contact interactions [7], making planning computationally expensive.



Fig. 1: Preliminary stacking results. Left: simulated structures for different target shapes. Right: a physical stone structure from an outdoor experiment.

Robotic bin-packing methods consider placement stability within containers [20, 9], where enclosing walls can provide lateral support unavailable in freestanding construction. More directly related, Johns et al. [12, 13] construct freestanding stone walls using contact-aware registration and simulation, while Liu and Napp [16] combine heuristic filtering with simulation-based assessments of structural strength to select placements. These stone-stacking methods do not use equilibrium across the entire structure to guide continuous pose refinement. Furthermore, they select placements using the current structure without explicitly comparing subsequent placement sequences, although masonry-inspired rules encourage suitable support for later objects. This leaves two challenges: refining poses according to contact-force redistribution throughout the structure and evaluating how current choices affect future feasible placements.

We address these challenges through pose optimization incorporating structural stability and GPU-accelerated Monte Carlo Tree Search (MCTS) for sequence planning.

Unlike prior stacking methods, we formulate placement refinement as a constrained bilevel problem and compute pose gradients of the objective and constraints using differentiable support functions (DSFs) for shape representation and collision detection [14, 2]. Bilevel formulations also appear in contact-aware manipulation planning and grasp generation [15, 21, 5]. Our lower-level problem accounts for force and torque equilibrium of all stones, including the candidate, with shared contact forces transmitting loads through the stack to the ground.

In addition, rather than relying solely on masonry heuristics,

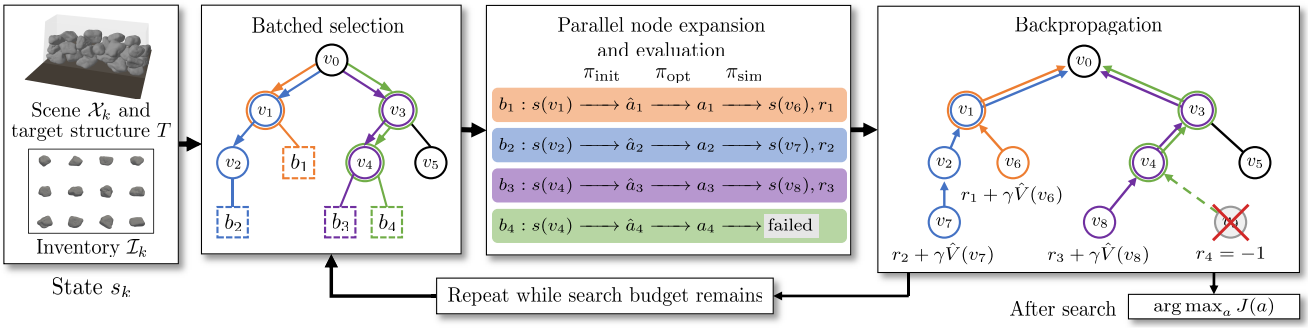


Fig. 2: Batched MCTS sequence planning. Colored requests are evaluated in parallel. Returns combine rewards and discounted leaf estimates. The dashed branch backpropagates its failure penalty once ($r_4 = -1$, $\hat{V}(v_9) = 0$) and is never selected again. After search, $J(a)$ ranks eligible root actions. See Sec. V-A for $\hat{V}(v)$ and $J(a)$.

we use MCTS to plan stacking sequences, explicitly accounting for how each placement affects subsequent placements. To adapt MCTS to the stacking problem, we design a hierarchical node expansion strategy that progressively refines and validates candidate placements through initial pose sampling, pose optimization, and simulation-based feasibility checking. We also define task-specific rewards that guide the search toward target shape coverage and structural stability.

However, repeated pose optimization and simulation in MCTS require large computational load. To make MCTS computationally practical, we introduce bodywise contact-force copies and an alternating direction method of multipliers (ADMM) decomposition for parallel force optimization within each candidate structure, while batching independent candidate placements on the GPU. Together with GPU-accelerated sub-ADMM simulation [11], this enables parallel candidate evaluation throughout the search tree (Figure 2) which significantly accelerate the overall MCTS process. Our main contributions are as follows:

- We formulate a differentiable bilevel placement problem that uses target geometry and force and torque equilibrium across the entire structure to guide candidate pose refinement.
- We adapt MCTS to stacking through hierarchical node expansion that samples, refines, and validates candidate placements, with task-specific rewards for target coverage and structural stability guiding lookahead over object choices and placement poses.
- We combine GPU-accelerated contact-force optimization and simulation for batched MCTS evaluation, using a bodywise ADMM decomposition to parallelize force optimization within each structure and evaluating independent candidate placements in parallel.

II. Related Works

A. Robotic Packing and Stone Stacking

Wang and Hauser [20] evaluate pile equilibrium in containers via linear programming, without using equilibrium residuals for pose refinement.

Johns et al. [12, 13] combine masonry rules, candidate contact-force optimization, registration, and simulation for freestanding wall construction, but pose updates do not account for load redistribution across the structure.

Liu et al. [17] filter simulation-settled poses geometrically; Liu and Napp [16] additionally evaluate structural strength through shake tests, without searching future placements. Menezes et al. [18] learn placement values reflecting future rewards with fixed object sequences and orientations. We instead combine whole-structure equilibrium-guided pose refinement with search over object choices and placement poses.

B. Contact-Aware Pose Optimization

Lee et al. [15] use differentiable contact factors and the envelope theorem for configuration inference after force optimization under a predefined task skeleton. GraspQP [21] refines grasps through KKT differentiation of a force-closure QP. BODex [5] couples bilevel grasp optimization with batched GPU contact-force solves using ReLU-QP. Its contact refinement uses finite differences or detached contact points, whereas we differentiate DSF contact geometry at regular solutions.

C. Simulation-Based Sequential Manipulation Planning

PMBS [8] combines parallel MCTS with GPU simulation for object retrieval from clutter. We adopt batched evaluation, adding parallel whole-structure contact-force optimization for pose refinement before simulation at each expansion.

III. Preliminaries and Overview

A. Shape Representation

We represent the target shape T and each object O_i as unions of smooth convex bodies modeled by differentiable support functions (DSFs) [14]. For a compact convex body $C \subset \mathbb{R}^3$, its support function is $h_C(x) = \max_{p \in C} x^\top p$ for $x \neq 0$. At pose $q = (R, t) \in \mathrm{SE}(3)$, we write

$$C(q) = \{Rp + t \mid p \in C\}, \qquad O_i(q_i) = \bigcup_j C_{ij}(q_i),$$

where C_{ij} are the convex components in object i's local frame. Similarly, $T = \bigcup_j T_j$, where T_j are fixed convex components in the world frame. DSFs support differentiable signed-distance and contact-geometry computations between posed convex bodies [2], enabling gradient-based pose optimization.

B. Markov Decision Process Formulation

Given an inventory of n objects, we model stacking as a finite-horizon sequential decision process in which one object is placed at each step. Let $\mathcal{P}_k \subseteq \{1, \dots, n\}$ be the indices of the objects already placed, and let $i_0 \notin \{1, \dots, n\}$ index the fixed ground. The current scene is $\mathcal{X}_k = \{(i, q_i) \mid i \in \mathcal{P}_k\}$, and the remaining inventory is $\mathcal{I}_k = \{1, \dots, n\} \setminus \mathcal{P}_k$. The state, action, and deterministic transition function are

$$s_k = (\mathcal{X}_k, \mathcal{I}_k, T), \quad a_k = (i_{k+1}, q_{k+1}), \quad s_{k+1} = F(s_k, a_k),$$

where $i_{k+1} \in \mathcal{I}_k$. The transition inserts the proposed pose into the scene, simulates the resulting structure, and returns the settled scene with $\mathcal{I}_{k+1} = \mathcal{I}_k \setminus \{i_{k+1}\}$. The reward $r(s_k, a_k)$ scores target-contained volume and physical quality; feasibility checks separately reject invalid transitions (Sec. V). The sequence planner therefore operates on the MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, F, r, \gamma)$.

C. Monte Carlo Tree Search

Monte Carlo Tree Search (MCTS) incrementally constructs a search tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$ over the MDP. Each node $v \in \mathcal{V}$ represents a state $s(v)$, and each edge $(v, a) \in \mathcal{E}$ represents an action and its transition to a child node. Let $N(v)$ denote the number of completed evaluations and $Q(v)$ the empirical mean of discounted returns backpropagated to that node.

Each iteration selects a path by balancing value and exploration, expands and evaluates an action, and backpropagates its discounted return to update Q and N.

In our planner (Fig. 2), expansion uses three modules: π_{init} proposes an object and initial pose $\hat{a}_k = (i_{k+1}, \hat{q}_{k+1})$, π_{opt} refines the pose, and π_{sim} evaluates the resulting structure to determine feasibility, the settled successor, and reward. The resulting flow is

$$s_k \xrightarrow{\pi_{\text{init}}} \hat{a}_k \xrightarrow{\pi_{\text{opt}}} a_k \xrightarrow{\pi_{\text{sim}}} s_{k+1},\ r(s_k, a_k).$$

Sec. V describes how these modules are integrated into batched search.

IV. Pose Optimization

The module π_{opt} refines $\hat{q}_{k+1}$ from π_{init} toward a stable, physically feasible placement q_{k+1}.

A. Single Object Pose Optimization Problem

Given s_k, we initialize q at $\hat{q}_{k+1}$ and solve the bilevel problem

$$\begin{aligned} \min_{q} \quad & \left[f_{\text{geo}}(q) + \min_{\lambda} f_{\text{phys}}(q, \lambda) \right] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \le 0, \quad j = 1, \cdots, n_c \end{aligned} \tag{2}$$

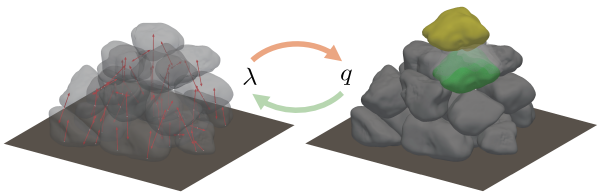


Fig. 3: Bilevel pose optimization. Left: optimizing contact forces λ under friction constraints minimizes structural equilibrium residuals at fixed q. Right: refining $q \in \text{SE}(3)$ minimizes the geometric and optimized physical objectives.

where $q \in \text{SE}(3)$ is the candidate object's pose and $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ collects all contact forces in the resulting structure, including inter-object and ground contacts. This formulation follows prior contact-aware optimization methods [15, 5].

The geometric objective f_{geo} is designed as follows:

$$\begin{aligned} f_{\text{geo}}(q) = w_{\text{h}} f_{\text{h}}(q) + w_{\text{in}} f_{\text{in}}(q \mid T, O_{i_{k+1}}) \\ + w_{\text{pen}} f_{\text{pen}}(q \mid \mathcal{X}_k, O_{i_{k+1}}) \end{aligned}$$

with positive weights. For $q = (R, t)$, $f_{\text{h}}(q) = t_z$ is the world-frame height of the candidate's pose origin. Writing $O = O_{i_{k+1}}$, we penalize proximity to existing objects and the fixed ground as

$$f_{\text{pen}}(q) = \sum_{j \in \mathcal{P}_k \cup \{i_0\}} \left[\epsilon_{\text{margin}} - \text{d}(O(q), O_j(q_j)) \right]_+^2,$$

where d is signed distance (positive for separation), $[x]_+ = \max(x, 0)$, and $\epsilon_{\text{margin}} \ge 0$ is the clearance margin. With $h_{O(q)}(u) = \max_j h_{C_{i_{k+1},j}(q)}(u)$, we define

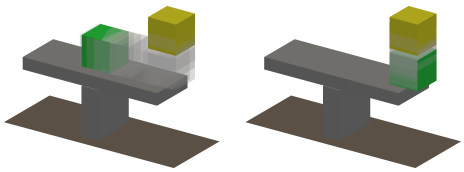
$$\begin{aligned} m_{\text{in}}(q) &= \max_j \min_{u \in \mathbb{S}^2} \left[h_{T_j}(u) - h_{O(q)}(u) \right], \\ f_{\text{in}}(q) &= \left[-m_{\text{in}}(q) \right]_+^2. \end{aligned}$$

This term penalizes violations of containment of the entire object within a single convex target component, a conservative condition for containment in T. Noting that the term is differentiable almost everywhere in q, we compute pose derivatives at regular inner solutions by implicit differentiation. Nonsmooth transitions have not caused practical difficulties in our implementation.

Let $\mathcal{P}_{k+1} := \mathcal{P}_k \cup \{i_{k+1}\}$ and let $I_l \subseteq \{1, \dots, n_c\}$ index all contacts involving object l, including ground contacts. The physical objective excludes the ground's equilibrium residual:

$$\begin{aligned} f_{\text{phys}}(q, \lambda) = w_{\text{eq}} \sum_{l \in \mathcal{P}_{k+1}} \| J_{c,I_l}^{\top}(q) \lambda_{I_l} + \lambda_{\text{ext},l} \|_W^2 \\ + w_{\text{cp}} \sum_{j \in I_{i_{k+1}}} \text{d}_j(q)^2 \|\lambda_j\|^2 + w_{\text{reg}} \sum_{j=1}^{n_c} \|\lambda_j\|^2 \end{aligned} \tag{3}$$

The three terms respectively penalize violations of static equilibrium, contact complementarity between the candidate and the current scene, and large contact forces through regularization. Figure 4 illustrates why equilibrium is evaluated across



(a) Structure-wide (b) Candidate-only

Fig. 4: Seesaw example: initial (yellow), optimized (green), and intermediate (gray) poses. Structure-wide equilibrium (a) moves the candidate toward the support; candidate-only equilibrium (b) leaves it near the unsupported end.

the structure rather than only for the candidate. The weight matrix $W \succ 0$ balances force and moment scales. Here, $\mathrm{d}_j(q)$ is the signed distance between the bodies of contact candidate j; J_{c,I_l}, $\lambda_{\mathrm{ext},l}$, and λ_{I_l} are object O_l's contact Jacobian, external wrench, and contact-force vector, respectively; and $w_{\mathrm{eq}}, w_{\mathrm{cp}}, w_{\mathrm{reg}} > 0$ are objective weights.

Finally, each $g_{\mathrm{cone},j}(\cdot)$ is the smoothed friction cone constraints for the j-th contact point defined by

$$g_{\mathrm{cone},j}(q,\lambda_j) = \sqrt{\|J^t_{c,j}\lambda_j\|^2 + \epsilon} - \mu_j J^n_{c,j}\lambda_j \tag{4}$$

where $\epsilon > 0$ is a smoothing parameter, μ_j is the friction coefficient, and $J^t_{c,j}$ and $J^n_{c,j}$ are the tangential and normal components of the Jacobian of the contact constraint.

B. Upper-Level Problem Optimization on SE(3) Manifold

Let $\lambda^*(q)$ be the solution of the lower-level problem and define $f_{\mathrm{phys}}(q) := f_{\mathrm{phys}}(q,\lambda^*(q))$. The envelope theorem [4] gives

$$\frac{df_{\mathrm{phys}}(q)}{dq} = \frac{\partial f_{\mathrm{phys}}(q,\lambda^*)}{\partial q} + \sum_{j=1}^{n_c} \nu_j^* \frac{\partial g_{\mathrm{cone},j}(q,\lambda_j^*)}{\partial q}$$

where ν_j^* is the optimal dual variable for the j-th friction cone constraint. Using the above gradient, we optimize the placement pose q on SE(3) manifold based on the Riemannian trust-region method [1] combined with BFGS Hessian approximation [10]. A homotopy schedule adjusts the geometric and physical weights between stages, holding them fixed within each stage and initializing from the preceding pose. It progressively reduces w_{reg} to limit regularization-induced equilibrium residuals. Candidate evaluation uses the final pose and forces.

C. Lower-Level Problem Optimization with ADMM

For fixed q, the lower-level problem is convex in λ. To apply bodywise ADMM [3], we introduce two local copies $\lambda_{j,1}$ and $\lambda_{j,2}$ per contact, one for each contacting body, including the ground. Collecting these copies into $\tilde{\lambda}$ makes each body's equilibrium residual depend only on its local variables. Consensus and friction constraints give the reformulation

$$\begin{aligned} \min_{\tilde{\lambda},z} \quad & \tilde{f}_{\mathrm{phys}}(q,\tilde{\lambda}) + \sum_{\substack{j=1,\dots,n_c \\ s=1,2}} \delta_{K_j}(z_{j,s}) \\ \text{s.t.} \quad & \lambda_{j,1} = \lambda_{j,2} = z_{j,1} = z_{j,2}, \qquad j = 1,\dots,n_c. \end{aligned}$$

where $\tilde{f}_{\mathrm{phys}}$ is the physical objective in local variables, $z_{j,s}$ are auxiliary variables, and δ_{K_j} is the indicator of $K_j := \{\lambda \in \mathbb{R}^3 \mid g_{\mathrm{cone},j}(q,\lambda) \le 0\}$. ADMM solves the body subproblems independently. An equivalent four-dimensional second-order cone (SOC) lifting enables closed-form cone projections for the friction updates, while consensus updates reduce to averaging.

The pose optimization module is implemented using CUDA to leverage GPU parallelization for efficient computation of the gradients and the ADMM updates.

V. Sequence Planning via Parallel MCTS

Building on Sec. III, we combine batched MCTS [8] with parallel pose optimization, sampling, simulation, and task-specific rewards (Fig. 2). We use a receding-horizon strategy: search considers a limited horizon of future placements, but only one selected placement is executed before replanning from the observed scene.

A. Batched Search Procedure

Batched Selection. Colored paths in the "Batched selection" panel (Fig. 2) terminate at expansion requests b_j. Over selectable child actions of v_p, we define

$$P(v_p,a) = \mathrm{softmax}_a(w_O \rho_O^{\mathrm{pre}}(a) - w_E e(a)),$$

where $\rho_O^{\mathrm{pre}}(a)$ is the pre-simulation object-volume fraction inside the target; $e(a)$ and the weights follow Sec. V-D. Writing $N_v = N(v)$ and similarly for $Q, \hat{N}$, we select evaluated children using PUCT [19]:

$$\mathrm{PUCT}(v) = \frac{N_v Q_v}{N_v + \hat{N}_v} + cP(v_p,a)\frac{\sqrt{N_{v_p} + \hat{N}_{v_p}}}{1 + N_v + \hat{N}_v}, \tag{5}$$

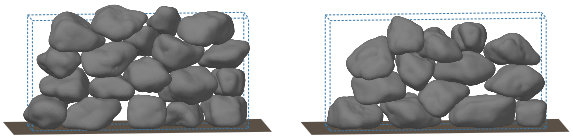
where $c > 0$ controls exploration, $\hat{N}$ counts temporary reservations, and the first term is zero when its denominator is zero. Non-root progressive widening [6] increases candidate counts with completed visits; reservations prevent duplicate batch evaluations without increasing this allowance.

Parallel Expansion and Evaluation. Each colored row in Fig. 2 evaluates one request through $\pi_{\mathrm{init}} \to \pi_{\mathrm{opt}} \to \pi_{\mathrm{sim}}$; rows run in parallel using the pose optimizer of Sec. IV. Selection descends only through evaluated successors and excludes failed candidates.

Leaf Value Estimation and Backpropagation. For non-terminal, feasible leaves, we replace rollouts with $\hat{V}(v) = \hat{r}\sum_{j=0}^{d_v-1}\gamma^j$, where d_v is capped by remaining search depth and inventory. The fixed hyperparameter $\hat{r}$ sets a heuristic per-step reward for unexplored continuations, influencing exploration through backpropagated returns alongside PUCT. Failed or terminal nodes have $\hat{V} = 0$. Following the arrows in the "Backpropagation" panel (Fig. 2), we propagate $r_v + \gamma\hat{V}(v)$, with incoming reward r_v, toward the root, adding discounted ancestor rewards, updating Q, N, and clearing reservations.

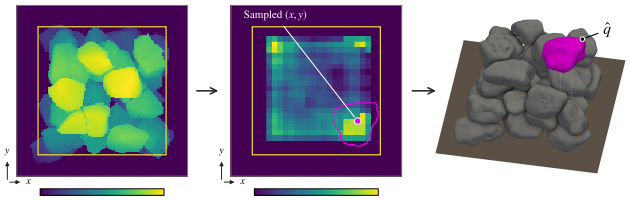
Action Selection and Tree Reuse. As shown at the bottom right of Fig. 2, we select the feasible root action maximizing

$$J(a) = (1-\beta)Q(v_a) + \beta G(a), \qquad \beta \in [0,1].$$



(a) MCTS (b) Greedy

Fig. 5: Wall construction using the same inventory of over 40 stones. MCTS places 19 stones (44.0% target filling); greedy stops at 14 (36.1%), finding no further feasible placement. Blue dashed outlines show the target.



(a) Scene height (b) Proposal score (c) Sampled initial pose

Fig. 6: Initial pose sampling for a fixed object and orientation: (a) scene height, (b) proposal score and sampled planar location, and (c) initial pose $\hat{q}$ above the scene.

Here $G(a)$ is the maximum discounted observed return over the deepest evaluated feasible paths below a, including its reward but excluding bootstrap values; root-only paths are eligible. This term offsets exploration failures that depress Q despite promising paths. If no root action is feasible, no placement is returned.

After executing the selected placement, we advance the root to the observed successor and repeat the search. Descendants and statistics are retained only when the observed state and planning context match the subtree; otherwise, the affected subtree is discarded.

Figure 5 compares the resulting planner with greedy placement.

B. Initial Pose Sampling

The sampler π_{init} selects objects with volume bias and uses finite orientation sets, seeding empty scenes near feasible target corners. For object i, orientation R, and feasible planar location $\mathbf{x}$, the score (Fig. 6) is

$$\sigma(\mathbf{x} \mid i, R) = w_{\text{s}}\sigma_{\text{s}} + w_{\text{f}}\sigma_{\text{f}} + w_{\text{b}}\sigma_{\text{b}} - w_{\text{h}}\sigma_{\text{h}} - w_{\text{d}}\sigma_{\text{d}},$$

with positive weights. The terms measure predicted bottom support fraction, unfilled layer area beneath the footprint, feasible-boundary proximity, normalized mean contact height, and a bounded footprint-gap penalty to the preceding object (zero for the first placement). Following Menezes et al. [18], floor and boundary preferences encourage filling empty regions and building inward from the boundary.

We combine elite selection with sampling without replacement weighted by $\exp(\sigma/\tau)$, $\tau > 0$. We initialize height so the candidate's lowest point lies above the base plane and the bounding-box tops of objects with overlapping padded XY bounds, with added clearance (Fig. 6(c)). Geometric filters constrain planar displacement and placement height after refinement and settling.

C. Simulation Module

At every search depth, π_{sim} first settles the candidate for a fixed duration with existing objects fixed, then resets velocities and simulates the entire structure. Settling requires a minimum duration and sustained low mass-normalized kinetic energy; excessive accumulated motion, invalid contacts, or timeout reject the placement.

The settled scene defines $s' = F(s, a)$. From rest in s', independent disturbance trials apply a force of magnitude $\alpha m_{i_{k+1}} g$ at the candidate's center of mass for Δt_f, in n_d evenly spaced horizontal directions ($\alpha > 0$, gravitational acceleration g). With all objects dynamic, every trial must recover within tolerances without excessive pose excursion; trials do not alter s'. GPU-accelerated subADMM [11] evaluates independent candidates in parallel.

D. Reward Design

For the new object $i = i_{k+1}$ at its settled successor pose q_i' (Sec. V-C), define

$$\rho_T = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(T)}, \quad \rho_O = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(O_i)}.$$

These voxel-estimated volume ratios reward target filling and containment. For feasible transitions, with fixed normalization scales absorbed into positive weights,

$$\begin{aligned} r(s,a) = {} & w_T\rho_T + w_O[\rho_O - \tfrac{1}{2}]_+ - w_E e(a) \\ & + w_H\big(\phi_H(s) - \phi_H(s')\big) - w_D D, \end{aligned}$$

where $\phi_H(s) = \text{Var}_\Omega(H_s)$ is the area-weighted variance of scene height over the target footprint Ω, with empty cells assigned zero height. Coverage lost through earlier objects' motion is not subtracted. Failed transitions receive $r = -1$.

For pose-optimization residual forces $\mathbf{f}_l$ and torques $\boldsymbol{\tau}_l$ across all placed objects,

$$e(a) = \max_l \left(\frac{\|\mathbf{f}_l\|}{m_l g} + \frac{\|\boldsymbol{\tau}_l\|}{m_l g \mathbb{V}(O_l)^{1/3}} \right).$$

Candidates with $e(a) \geq \epsilon_{\text{eq}}$ are rejected before simulation. Disturbance sensitivity measures motion from settled poses q_l' during force trials:

$$D = \min\left\{ D_{\max}, \max_{j,l,t} d_{\text{SE(3)}}\big(q_l^{(j)}(t), q_l'\big) \right\},$$

Here $q_l^{(j)}(t)$ is object l's pose in trial j at time t; the maximum spans all trials, objects, and simulation times including recovery, capped by $D_{\max}$. The distance $d_{\text{SE(3)}}$ sums translation distance and angular distance weighted by $\lambda_\theta > 0$. Feasibility rejection is independent of reward.

Acknowledgments

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